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Rich and rare ain't the common share. Describing biodiversity and informing conservation practices for South American anurans.

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Current claims of a biodiversity crisis demand immediate conservation actions. These require the identification of priority sites for conservation, which can be based on species richness patterns. Rarity, measured by the size of species' geographical ranges, is also frequently used as a single or complementary criterion. For instance, hotspots for conservation have been defined using either one or the other criterion. Here we use a new and simple tool, range-diversity plots that simultaneously analyze species richness and range size, to identify a set of richness-rarity sites. We applied this tool to the anurans of South America and show how it can be readily used to inform conservation practices. Also, we evaluated the representation of these sites within the protected area system. We built a presence-absence matrix from the anuran range maps overlaid by a 1°-resolution grid and recorded the number of species and the average range size for assemblages at each cell. South American anurans show congruent patterns between richness and rarity, with rich-rare sites harboring high numbers of species averaging small range sizes. Nonetheless, there are also poor-rare sites that contain few species averaging small range sizes. Both sets of sites are poorly represented in the protected area system. We show that the South American protected area system can be improved by considering patterns of richness and rarity simultaneously, which can be easily described for a quick assessment of biodiversity at geographic scales and to compare against and contribute to the definition of conservation priorities.